

Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 8: Winds, Storms and Cyclones

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	8 — Winds, Storms and Cyclones
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Traps target the warm-air-is-heavier and hot-air-rises-by-a-special-force errors, the fast-air-pushes-outward (Bernoulli) reversal, the can-crush 'cold air is heavier / metal folds itself' misreads, the wind-blows-low-to-high direction flip, the land-vs-sea heating-rate confusion in breezes and monsoon, the 'eye is the most dangerous part' and 'high-pressure eye' cyclone mistakes, the tornado-bigger-than-cyclone scale error, the sound-faster-than-light thunder mix-up, the storm-surge-is-just-rainwater understatement, and the barometer/anemometer/thermometer instrument swaps. Do not write on this paper — mark answers on your answer sheet.

1. A sealed glass jar of air is carried from a valley up to a mountain peak where the outside air is thinner. The amount of air trapped inside the jar is unchanged. Compared with the valley, the OUTWARD push of the trapped air on the jar's walls is:

- (A) greater, because high places have stronger air pressure
- (B) zero, because air loses all pressure on mountains
- (C) unchanged, because the trapped air is the same — but the inward push from the now-thinner outside air is less, so the jar feels a net outward strain
- (D) less, because the trapped air also thins out on the mountain

2. In the paper-strip activity you blow hard across the TOP of a strip held below your lip, and it lifts. A student claims it lifts because 'your breath pushes the paper up from above.' The real reason it lifts is:

- (A) your breath physically scoops under the strip and lifts it
- (B) fast-moving air above the strip has lower pressure, so the still air below — at higher pressure — pushes the strip up
- (C) warm breath is lighter than the paper and floats it
- (D) blowing removes the weight of the paper

3. In the can-crushing demo, a little water is boiled in an open can so steam drives most of the air out, then the can is sealed and cooled. Which single change is most directly responsible for the crushing?

- (A) the cold air outside becomes heavier and physically heavier-than-before
- (B) the metal of the can contracts so violently that it folds itself
- (C) cooling produces extra water that pushes the walls inward
- (D) the steam inside condenses back to a tiny drop of water, leaving very little gas, so the inside pressure falls far below the outside pressure

4. Two thin sheets of paper hang a few centimetres apart. You blow a stream of air down the gap between them. Many students predict they fly apart. They actually swing TOGETHER because:

- (A) your breath glues the sheets together with moisture
- (B) the air in the gap pushes outward harder than the outside air
- (C) the moving air in the gap is at lower pressure, so the still higher-pressure air on the outer faces pushes both sheets inward
- (D) blowing makes both sheets lighter so gravity pulls them in

5. Why are we, walking around at sea level, not crushed by the atmosphere pressing on us from every direction?

- (A) the atmosphere actually exerts no real pressure on a human body
- (B) the fluids and air inside our bodies push outward with the same pressure, so the inward and outward pushes balance
- (C) our skin is rigid and completely seals air out
- (D) gravity pulls the air pressure straight down and away from us

6. A balloon is stretched over a bottle's mouth and the bottle is dipped in HOT water; the balloon inflates. The bottle is then moved into ICE water. Predict the balloon and give the reason:

- (A) it inflates even more, because cold air takes up more room
- (B) it deflates and gets sucked inward, because the cooled air contracts and its pressure drops below the outside pressure
- (C) it stays exactly as inflated as it was in hot water
- (D) it pops off, because cold air pushes outward harder

7. An inflated balloon is left in a cold car overnight and looks shrunken by morning, then re-swells when brought into a warm room. No air leaked out. The best explanation is:

- (A) some air leaked out in the cold and leaked back in when warm
- (B) cold destroys part of the air and warmth re-creates it
- (C) cooling makes the trapped air contract (take less space); warming makes it expand again, with the same air all along
- (D) the rubber itself shrinks and grows but the air is unchanged

8. A candle is lit under one end of a balanced, pivoted rod that has an open paper bag hung at each end. The bag over the flame slowly rises. This happens because the air inside that bag:

- (A) is heated, expands and becomes less dense (lighter), so it is pushed up by the cooler, denser air around it
- (B) gains weight from the smoke and tips downward then up
- (C) is cooled by the flame and sinks, lifting the other side
- (D) loses all its air to the flame and floats empty

9. Above a sun-baked parking lot the air rises, and a gentle breeze is felt moving in along the ground toward the lot. Put the cause and effect in correct order:

- (A) a breeze starts first → it heats the lot → then the air rises
- (B) the air rises first for no reason → this cools the lot → then a breeze forms
- (C) cooler air sinks onto the lot → it warms → then it stays put with no breeze
- (D) the lot heats the air → that air expands, becomes less dense and rises → pressure near the ground falls → cooler surrounding air flows in as a breeze

10. Wind always blows from a high-pressure region toward a low-pressure region. Region X has higher pressure than region Y, and region Y has higher pressure than region Z. The strongest wind, and its direction, is most likely:

- (A) between X and Y, blowing from Y toward X
- (B) between Y and Z, blowing from Z toward Y
- (C) there is no wind because all three pressures exist at once
- (D) between X and Z, blowing from X toward Z, because that pair has the largest pressure difference

11. A weather map shows two neighbouring areas, one marked HIGH pressure and one marked LOW pressure, very close together with a steep change between them. Compared with a day when the same two areas have nearly equal pressure, today the wind between them will be:

- (A) weaker, because high pressure holds the air still
- (B) stronger, because a larger pressure difference over a short distance drives faster air movement
- (C) exactly the same, because wind speed does not depend on pressure
- (D) absent, because air only moves when pressures are equal

- 12.** Why does cooler air, not warmer air, move in to take the place left by air that has risen above a hot field?
- (A) cooler air is lighter than warm air, so it floats over to the gap
 - (B) the rising warm air leaves lower pressure behind it, and the denser, higher-pressure cooler air nearby flows in to fill that gap
 - (C) the Sun directly pushes the cool air sideways into the gap
 - (D) warm air is attracted back down into the same spot it left
- 13.** Hot air is sometimes described as 'lighter' than cold air. The most accurate meaning of this is that, for the same volume, hot air:
- (A) contains fewer kinds of gases than cold air
 - (B) has lost some of its matter to the Sun
 - (C) weighs less because heating made it expand, spreading the same amount of air over more space (lower density)
 - (D) is pulled upward by a special force only warm air feels
- 14.** At a coast on a hot afternoon, the wind blows from sea to land. The KEY property that makes this happen is that, under the same sunshine:
- (A) water heats up faster than land, so air rushes from the land to the sea
 - (B) land and sea heat at exactly the same rate, so the wind is random
 - (C) land heats up faster than water, so the air over land rises and lower-pressure forms there, drawing in cooler air from over the sea
 - (D) the sea is always hotter than the land in the afternoon
- 15.** At night a coastal wind reverses and blows from land to sea (land breeze). The reason the direction flips after sunset is:
- (A) land cools faster than the sea, so now the warmer air is over the sea, rises there, and draws air outward from the land
 - (B) the sea cools faster than the land, so air rushes from the sea onto the land
 - (C) winds simply reverse at sunset everywhere for no physical reason
 - (D) the land stays warmer than the sea all through the night
- 16.** A coastal town normally gets a strong afternoon sea breeze. On one cloudy, cool afternoon the land fails to heat up and is actually slightly cooler than the sea. That afternoon you should expect:
- (A) an even stronger sea breeze than usual
 - (B) the wind to blow straight up with no horizontal direction
 - (C) the sea breeze to weaken or even reverse, because the pressure difference that drives it is gone or flipped
 - (D) no change at all, since sea breezes do not depend on temperature
- 17.** Coastal cities often have milder days and nights than far-inland cities at the same latitude. The best reason, using this chapter's ideas, is that:

- (A) the sea has no effect, and coastal mildness is pure coincidence
- (B) water heats and cools more slowly than land, so day and night breezes between sea and coast even out the temperature there
- (C) inland air has no weight, so it cannot hold heat
- (D) the coast simply receives far less sunlight than inland areas

18. India's summer monsoon brings heavy rain because, in summer, the land heats more than the sea. This drives the monsoon winds to blow:

- (A) from the land toward the sea, carrying dry air away
- (B) from the sea toward the land, carrying moisture that falls as rain
- (C) from the poles straight to the equator, bringing cold dry air
- (D) straight downward out of the upper sky

19. During the WINTER months over much of India the prevailing winds tend to blow from the land toward the sea. Compared with the summer monsoon, these winter winds are mostly:

- (A) even wetter, because cold winds carry more rain
- (B) identical in moisture to the summer monsoon
- (C) always blowing straight up, so they bring snow
- (D) dry, because they come off the cooler land rather than picking up moisture from the sea

20. The whole large-scale pattern of winds on Earth, including the monsoon, is ultimately set up by:

- (A) the Moon's gravity pulling the air around the globe
- (B) ocean currents acting entirely on their own, with no role for the Sun
- (C) the combined spinning of household fans and machines
- (D) the uneven heating of the Earth's land, sea and air by the Sun

21. The Sun heats air over the equator much more than air over the poles. The direct large-scale consequence of this difference is that:

- (A) warm equatorial air rises and cold polar air sinks, setting up giant wind currents that carry heat around the planet
- (B) the poles become hotter than the equator over time
- (C) all wind on Earth stops because the temperatures balance
- (D) air near the equator permanently loses its weight

22. Thunderstorms are common in hot, humid weather but rare on cold, dry days. The reason hot humid air is so favourable is that it:

- (A) is warm enough to rise rapidly and carries plenty of water vapour that can condense into towering storm clouds
- (B) is too heavy to move, so it piles up and explodes into a storm

- (C) contains no water, so the clouds form from dry dust instead
- (D) is cooler than the ground, so it sinks and triggers lightning below

23. Inside a developing thunderstorm, which combination is the chief engine of its violence?

- (A) cold dry air quietly sinking to the ground with no moisture
- (B) rapidly rising warm, moist air together with water droplets and ice falling back down through it
- (C) lightning forming first and then dragging the air upward
- (D) the ground touching and heating the cloud from below

24. We see a lightning flash and hear its thunder a few seconds later, even though both are produced at the same instant. The delay tells us that:

- (A) the thunder is actually created several seconds after the flash
- (B) light travels much faster than sound, so the flash reaches us almost at once while the sound lags behind
- (C) sound travels faster than light, so it should arrive first
- (D) our eyes react seconds faster than our ears

25. During a thunderstorm, lightning is best described as:

- (A) a huge electric spark caused by electric charges that build up in the storm clouds
- (B) sunlight reflected brightly off the falling rain
- (C) burning gas catching fire high in the sky
- (D) thunder's sound turning itself into a flash of light

26. If you count the seconds between seeing lightning and hearing the thunder, a SHORTER gap each time tells you that:

- (A) the storm is moving away, because closer storms have longer gaps
- (B) the lightning is getting weaker, not nearer
- (C) the speed of light is increasing during the storm
- (D) the storm is getting closer, because the sound has less distance to travel before it reaches you

27. As a cyclone forms over a warm sea, water vapour in the rising air condenses into droplets. This condensation FEEDS the storm because it:

- (A) cools the air sharply and brings the whole storm to a halt
- (B) releases heat that warms the air further, lowering the central pressure even more and pulling in faster wind
- (C) makes the air heavier so it sinks back down and weakens the storm
- (D) raises the pressure at the centre, slowing the wind

28. Arrange the self-feeding cycle of a growing cyclone in the correct order: (i) faster wind rushes in toward the centre; (ii) the Sun warms the sea and warm moist air rises; (iii)

vapour condenses, releasing heat and lowering the pressure further.

- (A) (i) → (ii) → (iii)
- (B) (iii) → (ii) → (i)
- (C) (ii) → (i) → (iii)
- (D) (ii) → (iii) → (i)

29. A cyclone strengthens over the warm open sea but quickly weakens after it moves over land. The main reason for the weakening is that over land it:

- (A) is colder than outer space and freezes solid
- (B) is simply blocked by buildings acting as a wall
- (C) loses its eye the instant it touches soil, which stops the wind
- (D) is cut off from the warm, moist sea air that supplied the heat and vapour feeding it

30. From the centre of a cyclone outward, which sequence of features is correct?

- (A) a violent eye → a calm eye wall → a still outer ring
- (B) a high-pressure eye → a dry eye wall → no clouds at all
- (C) a calm, low-pressure eye → a violent eye wall of fastest winds and heaviest rain → outer bands of spiralling clouds
- (D) a calm eye → a calm eye wall → a calm edge with no wind anywhere

31. A ship is unlucky enough to sail right through the centre of a cyclone. The crew would most likely experience, in order:

- (A) steady mild wind the whole way through with no change
- (B) no wind at all from start to finish
- (C) fierce wind, then a sudden eerie calm as the eye passes over, then fierce wind again from the opposite direction
- (D) calm, then violent wind, then a permanent calm

32. As a cyclone INTENSIFIES over the sea, what happens to its central pressure and its wind speed?

- (A) the central pressure rises and the winds blow faster
- (B) the central pressure falls and the winds blow slower
- (C) the central pressure falls lower and the winds blow faster
- (D) both the central pressure and the wind speed stay constant

33. A barometer at a coastal station shows the air pressure falling unusually fast over a few hours. The most reasonable interpretation is:

- (A) a storm or cyclone may be approaching, because such systems are regions of low pressure
- (B) clear, calm, settled weather is on the way
- (C) the wind is about to stop completely
- (D) the air temperature has just dropped below freezing

34. The very low pressure at the centre of a cyclone is crucial because it:

- (A) draws the surrounding higher-pressure air rushing inward, which becomes the cyclone's fierce winds
- (B) keeps the air around it perfectly still
- (C) raises the sea level only far out at sea and nowhere near the coast
- (D) dries out and warms the upper air so no clouds can form

35. The same kind of violent tropical storm is called a 'cyclone' near India but a 'hurricane' or 'typhoon' elsewhere. The best conclusion is that these are:

- (A) three completely different and unrelated weather events
- (B) names for cyclones of three fixed, different sizes
- (C) older names for what are now only called tornadoes
- (D) the same type of storm given different regional names in different parts of the world

36. A spinning weather system has these features: a roughly circular shape, a quiet low-pressure centre ringed by the most violent winds, swirling cloud bands, and a diameter of hundreds of kilometres. This is best identified as a:

- (A) cyclone
- (B) tornado
- (C) land breeze
- (D) thunderstorm with no rotation

37. Along a low coast, the single greatest cause of death and destruction when a cyclone makes landfall is usually the:

- (A) loud thunder echoing across the shore
- (B) storm surge — a huge mound of sea water driven inland that floods the low coast
- (C) hailstones falling from the storm clouds
- (D) bright lightning flashes over the water

38. Why does a cyclone's storm surge flood low-lying coasts so severely?

- (A) the surge is only slow rainwater that gradually collects on the coast
- (B) the sea level drops, exposing dry land that then floods
- (C) the surge is made entirely of fresh river water flowing backward
- (D) the very low central pressure and fierce winds heap up a great mound of sea water that is driven far inland

39. Coastal communities suffer far fewer cyclone deaths today than a century ago. The biggest single reason is that:

- (A) cyclones themselves have become much weaker over the years
- (B) the sea no longer floods the coast at all
- (C) weather satellites and radar now give accurate warnings a day or two ahead, so people can move to shelters in time
- (D) people have completely stopped living anywhere near the coast

40. When an official cyclone warning is issued for your coastal town, the SAFEST immediate response is to:

- (A) go to the beach to watch the giant waves
- (B) move to a strong shelter and follow the official instructions
- (C) ignore it until the wind is already strong
- (D) drive toward the sea to beat the inland traffic

41. Which of these is a sound preparedness measure once a cyclone warning is in force?

- (A) throw all the doors and windows wide open to let the wind blow through
- (B) go outside to take photographs of the approaching storm
- (C) store clean drinking water and keep emergency supplies and a battery radio ready
- (D) shelter under a tall, lone tree in the open

42. Before a cyclone, fishermen are told not to go to sea chiefly because:

- (A) the violent winds and high waves can capsize and sink their boats
- (B) the fish all swim away whenever a storm is near
- (C) sea water turns too cold to sail in during a cyclone
- (D) a cyclone makes the sea water vanish, stranding the boats

43. A train is about to be passed at high speed by an express. A person standing very close to the platform edge can be dangerously pulled toward the moving train. The pressure reason is that:

- (A) the fast-moving air between the person and the train is at lower pressure, so the higher-pressure air behind the person pushes them toward the train
- (B) the moving train blows air that pushes the person toward it
- (C) the train magnetically attracts the person standing nearby
- (D) the rails vibrate and physically drag the person forward

44. A light ball hovers steadily in the upward air stream from a vertical hair-dryer and even springs back when nudged sideways out of the stream. The ball stays IN the stream because:

- (A) the air in the stream is sticky and glues the ball in place
- (B) the ball becomes weightless once it is inside the stream
- (C) the dryer's heat lifts the ball by a magnetic pull
- (D) the fast-moving air in the stream is at lower pressure, so the slower, higher-pressure air around it pushes the ball back into the stream

45. A stiff card is held over a brim-full upturned glass of water and the water stays in. The card and water do not fall because:

- (A) the water is glued to the card by being wet
- (B) water simply cannot fall when it is upside down
- (C) the glass holds the water in by a magnetic force
- (D) the upward push of air pressure on the card is greater than the downward weight of the water

46. Two identical sealed tins, one taken to a high mountain top and one left at the seashore, are compared. The mountain-top tin feels a different outside air pressure because:

- (A) there is more air above it on the mountain, so the pressure is higher
- (B) the mountain air has no pressure at all
- (C) the pressure is exactly the same everywhere on Earth
- (D) there is less air above it on the mountain, so the outside air pressure pushing on it is lower

47. On a hot day the air just above sun-baked ground tends to rise. Choosing the correct chain of reasoning:

- (A) the ground cools the air → it shrinks and becomes heavier → so it rises
- (B) the ground heats the air → the air expands and becomes less dense → being lighter than the cooler air around it, it is pushed up
- (C) the air turns into water → and the water vapour lifts it
- (D) the air gains weight from the heat → and heavy air always rises

48. A region of the atmosphere is being strongly warmed from below for several hours. Over that time the region tends to develop:

- (A) lower pressure, because the warmed air expands and rises away, leaving less air pressing down
- (B) higher pressure, because warm air is heavier and presses down more
- (C) no change in pressure, because heating does not affect pressure
- (D) zero pressure, because warmth removes all the air

49. Knowing that by day the air over warm land is at LOWER pressure than over the cooler sea, and that wind blows from high to low pressure, the daytime coastal breeze must blow:

- (A) from the land toward the sea
- (B) from the sea toward the land
- (C) straight upward only, with no sideways direction
- (D) round and round in a circle with no net direction

50. Why does a cyclone keep drawing in more and more air as it grows, instead of settling into a calm?

- (A) the wind itself manufactures new low pressure with no need for heat or moisture
- (B) the calm eye sucks air in by a vacuum cleaner effect unrelated to pressure
- (C) condensation in the rising air keeps releasing heat and lowering the central pressure, so air keeps rushing in toward that ever-deeper low
- (D) the storm cools the sea, and cool seas always pull in wind

51. A sealed plastic bottle of warm air is capped and put in a freezer; an hour later it is dented inward. The dent appears because:

- (A) the freezer mechanically squeezed the bottle from outside
- (B) the cooled air inside contracted and its pressure dropped, so the higher outside air pressure pressed the sides in
- (C) cold made the plastic itself expand and fold inward
- (D) water from the freezer soaked through and pulled the walls in

52. A region's air pressure is described as 'high.' Compared with a nearby 'low' region, the air in the high-pressure region tends to:

- (A) flow outward toward the low-pressure region, since wind moves from high to low pressure
- (B) stay completely trapped and never move toward the low region
- (C) flow inward from the low region, since air moves from low to high
- (D) rise straight up and leave the surface empty of air

53. Why does warm air rising over a hot region create a region of LOWER pressure at the surface beneath it?

- (A) rising air leaves more air behind, raising the pressure
- (B) the Sun adds extra pressure to cool surfaces only
- (C) as the warm air rises and moves away, there is less air left pressing down on the surface, so the pressure there drops
- (D) rising air turns into water that weighs the surface down

54. A demonstration shows two ping-pong balls on threads swinging together when air is blown between them, just like ships sailing close in parallel can be drawn together. The shared principle is:

- (A) fast-moving air (or water) exerts lower pressure, so the higher pressure on the outer sides pushes the objects together
- (B) moving fluids exert higher pressure, so they push the objects apart
- (C) moving air carries the objects along by sticking to them
- (D) the objects attract each other by a magnetic force when fluids move

55. If the central pressure of one cyclone is much lower than that of another over the same warm sea, you can reasonably expect the lower-pressure cyclone to have:

- (A) slower winds, because lower pressure calms the air
- (B) no wind, because the air has nowhere to go
- (C) faster winds, because a deeper central low pulls in surrounding air more strongly
- (D) wind that blows outward from the centre

56. A teacher heats the air in a flask connected by a tube to coloured water in a beaker, and the water level in the tube falls as the flask warms, then rises again as the flask cools. This shows that air:

- (A) is created when heated and destroyed when cooled
- (B) always pushes water upward whether heated or cooled

- (C) has no response at all to changes in temperature
- (D) expands when heated (pushing the water down) and contracts when cooled (letting the water rise back)

57. A student says: 'Wind blows from cold places to hot places.' Using the chapter's pressure idea, the most precise correction is:

- (A) the student is completely right because cold air is always attracted to heat
- (B) wind blows from hot to cold, the exact opposite, in every case
- (C) wind direction has nothing to do with either temperature or pressure
- (D) wind blows from higher pressure to lower pressure; since hot regions usually become low-pressure areas, the wind often does end up moving from cooler toward warmer regions — but pressure, not temperature, is the true driver

58. In the can-crush demo, why is it important to BOIL the water first and drive out the air, rather than just sealing a can of room-temperature air and cooling it?

- (A) boiling fills the can with steam that later condenses to almost nothing, giving a far bigger pressure drop and a dramatic crush
- (B) boiling makes the metal soft so any cooling will crush it
- (C) room-temperature air cannot be cooled at all
- (D) boiling adds extra air that strengthens the crush

59. Why does sealing the can in the crush demo matter — what would happen if a large hole were left open while it cooled?

- (A) outside air would flow in through the hole to equalise the pressure, so no pressure difference would build and the can would not crush
- (B) the can would crush even faster because air escapes more easily
- (C) the metal would melt through the open hole
- (D) the steam would catch fire at the opening

60. Why can the SAME cyclone bring fierce wind, a period of eerie calm, and then fierce wind again to one coastal town as it passes directly overhead?

- (A) the cyclone stops and restarts itself twice over the town
- (B) two separate cyclones happen to arrive one after another
- (C) the leading eye wall brings fierce wind, the calm eye passes over the town, then the trailing eye wall brings fierce wind from the opposite side
- (D) the wind simply changes its mind for no physical reason

Solutions — Science (Class 7), Chapter 8: Winds, Storms and Cyclones — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	A	31	C	41	C	51	B
2	B	12	B	22	A	32	C	42	A	52	A
3	D	13	C	23	B	33	A	43	A	53	C
4	C	14	C	24	B	34	A	44	D	54	A
5	B	15	A	25	A	35	D	45	D	55	C
6	B	16	C	26	D	36	A	46	D	56	D
7	C	17	B	27	B	37	B	47	B	57	D
8	A	18	B	28	D	38	D	48	A	58	A
9	D	19	D	29	D	39	C	49	B	59	A
10	D	20	D	30	C	40	B	50	C	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) The air sealed inside is the same air at the same temperature, so the push it exerts on the walls does not change. What changes is outside: the mountain air is thinner and pushes inward less, so the balance shifts and the walls feel a net outward strain. 'Greater because high places have stronger pressure' is backwards — air pressure is lower at altitude. 'Zero' is false; the trapped air still has the same pressure. 'Less, because the trapped air thins out' is wrong — the jar is sealed, so the trapped air's amount and pressure cannot change just by moving it.

2. (B) Fast-moving air exerts lower pressure. Blowing across the top makes the pressure above the strip drop, so the higher-pressure still air underneath pushes the strip up. The student's claim is wrong because air blown OVER the top cannot push it up from above — that would push it down. 'Breath scoops under' is wrong; you blow across the top, not underneath. 'Warm breath floats it' is irrelevant — the effect works with any fast stream regardless of warmth. 'Removes the weight' is meaningless — weight is unchanged.

3. (D) Boiling fills the can with steam and pushes air out; sealing and cooling makes that steam condense to a few drops of liquid, leaving almost no gas inside. The inside pressure collapses while the outside air pressure stays high, so the atmosphere crushes the can. 'Cold air becomes heavier' is not the mechanism — the imbalance is between inside and outside pressure, not air weight. 'Metal folds itself' is wrong; metal contraction is far too small to crush a can. 'Extra water pushes the walls in' is backwards — water forming reduces the inside pressure, it does not push outward.

4. (C) Moving air has lower pressure than still air. The stream lowers the pressure in the gap, so the still air pressing on each sheet's outer face — now at higher pressure — pushes the two sheets toward each other. The 'fly apart' prediction wrongly assumes the moving air pushes outward. 'Breath glues them' is irrelevant to the pressure effect. 'Gap air pushes harder outward' is the exact opposite of what fast air does. 'Lighter so gravity pulls in' makes no sense — gravity acts downward, not inward.

5. (B) Air pressure pushes inward from all sides, but the fluids and gases inside our bodies push outward with an equal pressure, so the two balance and we feel nothing. 'No real pressure' is false — the same atmosphere crushes a sealed can. 'Skin is rigid and seals air out' is wrong; skin is flexible and the balance is about internal pressure, not sealing. 'Gravity pulls pressure away' confuses gravity with the all-directional push of air pressure.

6. (B) Hot water expanded the air and inflated the balloon. Ice water reverses this: the air contracts and its pressure falls below the outside atmosphere, so the balloon is pushed inward, even sucked into the neck. 'Inflates more' is wrong — cooling shrinks air, it does not expand it. 'Stays the same' ignores that the temperature changed. 'Pops off because cold air pushes harder' is backwards — cold air pushes LESS, not more.

7. (C) Air expands when heated and contracts when cooled. The cold shrank the trapped air so the balloon looked smaller; warming let it expand again — the same air throughout. 'Air leaked out and back in' contradicts 'no air leaked.' 'Cold destroys air' is false; matter is not destroyed by cooling. 'Only the rubber shrinks' is wrong — rubber's size change is tiny; the visible change is the air's volume responding to temperature.

8. (A) Heated air expands; the same air now spreads over more space, so each litre is lighter (less dense). The surrounding cooler, denser air sinks under it and pushes the warm bag up. 'Gains weight from smoke' is wrong — warm air becomes lighter, not heavier. 'Cooled by the flame' is contradictory; a flame heats. 'Loses all its air' is nonsense — the air is heated, not removed.

9. (D) The hot surface heats the air, which expands, grows less dense and rises; this leaves a region of lower pressure near the ground, so cooler surrounding air flows in — that inflow is the breeze. The other orderings put effects before their causes: a breeze cannot start before the heating that drives it; air does not rise 'for no reason'; and cooler air sinking onto the lot with 'no breeze' ignores that the rising hot air must be replaced.

10. (D) Wind blows from high to low pressure, and a bigger pressure difference drives faster wind. Since X is highest and Z is lowest, the X-to-Z gap is the largest, giving the strongest wind from X toward Z. 'Y toward X' blows from lower to higher pressure — impossible. 'Z toward Y' also blows from low to high — impossible. 'No wind' is wrong; differences in pressure are exactly what create wind.

11. (B) The larger and steeper the pressure difference, the faster the air rushes from high to low, so the wind is stronger today. 'High pressure holds air still' is wrong — high pressure is the source the air flows FROM. 'Wind does not depend on pressure' contradicts the core idea of the chapter. 'Air moves only when pressures are equal' is exactly backwards — equal pressures mean no wind.

12. (B) Rising warm air leaves a low-pressure gap; the surrounding cooler air is denser and at higher pressure, so it flows in to fill the gap — that inflow is wind. 'Cooler air is lighter' is false; cool air is denser and heavier than warm air. 'Sun pushes air sideways' is not the mechanism — the pressure difference moves the air. 'Warm air comes back down' contradicts the fact that warm air keeps rising.

13. (C) When air is heated it expands, so a given volume now holds less air than the same volume of cold air — it is less dense, hence 'lighter.' 'Fewer kinds of gases' is wrong; heating does not change which gases are present. 'Lost matter to the Sun' is false; heating adds energy, not removes matter. 'A special upward force only warm air feels' is a misconception — warm air rises because the denser cool air around it sinks and displaces it.

14. (C) Land warms faster than water under the same Sun. The air over the warmer land rises, leaving lower pressure there, so cooler higher-pressure air over the sea moves in — a sea breeze. 'Water heats faster' is the opposite of the truth. 'Same rate, random wind' is wrong — the different heating rates are exactly what set the direction. 'Sea always hotter' is false; by afternoon the land is hotter, which is why the breeze comes off the sea.

15. (A) Land loses heat faster than water, so at night the sea is the warmer surface. Air rises over the warmer sea, lowering pressure there, and cooler air flows out from the land — a land breeze, the reverse of daytime. 'Sea cools faster' is wrong; water holds heat longer. 'Reverse for no reason' ignores the cause — the temperature roles swap. 'Land stays warmer all night' is false; land cools quickly after sunset.

16. (C) The sea breeze depends on the land being hotter than the sea. If the land fails to heat and is cooler than the sea, the usual pressure difference vanishes or reverses, so the breeze weakens or flips toward the sea. 'Even stronger' is wrong — the driving difference is weaker, not stronger. 'Straight up' ignores that breezes are horizontal flows driven by horizontal pressure differences. 'No change, temperature doesn't matter' contradicts the whole mechanism of breezes.

17. (B) Water changes temperature slowly, so the nearby sea acts as a buffer; daytime sea breezes cool the coast and night-time land breezes are mild, smoothing out extremes. 'Sea has no effect / coincidence' ignores the breeze mechanism. 'Inland air has no weight' is false — all air has weight. 'Coast gets far less sunlight' is wrong; sunlight is similar at the same latitude, the difference is water's slow heating.

18. (B) In summer the land becomes hotter than the sea, so air over the land rises and lower pressure forms there; moisture-laden air flows in from the sea toward the land and its water falls as monsoon rain. 'Land to sea' describes the dry winter direction, not the rainy summer monsoon. 'Poles to equator, cold dry air' is not the monsoon and would not bring heavy rain. 'Straight downward' is not how winds blow.

19. (D) Winter winds blow from land to sea; having travelled over land, they carry little moisture and are mostly dry — the opposite of the moisture-laden summer monsoon. 'Even wetter, cold winds carry more rain' is wrong; land-origin winter winds are dry. 'Identical in moisture' ignores that direction (and origin over land vs sea) determines moisture. 'Straight up bringing snow' misdescribes the horizontal, dry winter flow.

20. (D) Because the Sun heats the equator far more than the poles, and land differently from sea, the resulting pressure differences set up the planet's wind patterns, including the monsoon. 'Moon's gravity' drives tides, not the global winds. 'Ocean currents alone' is wrong — the currents themselves and the winds both trace back to solar heating. 'Household fans' is absurd at a planetary scale.

21. (A) The equator's strong heating makes its air rise; cold dense polar air sinks, and air flows between them — giant currents that move heat across the globe and drive the prevailing winds. 'Poles hotter than equator' reverses reality. 'All wind stops' is wrong — the temperature difference is what CREATES wind. 'Air loses its weight' is false; air always has weight.

22. (A) Hot air rises quickly and humid air supplies the water vapour; as the rising air cools, that vapour condenses into the tall clouds of a thunderstorm. 'Too heavy to move' is wrong — warm air is light and rises. 'Contains no water' contradicts 'humid.' 'Cooler than the ground and sinks' is backwards — the storm air is warm and rises, it does not sink.

23. (B) A thunderstorm is driven by strong up-currents of warm moist air meeting droplets and ice falling down — these violent up-and-down motions build charge and turbulence. 'Cold dry sinking air' would suppress, not power, a storm. 'Lightning first, dragging air up' reverses cause and effect — the rising air comes first and helps make the lightning. 'Ground heating the cloud by touch' is not how the high cloud is energised.

24. (B) Light travels far faster than sound, so the flash arrives almost instantly while the slower sound (thunder) reaches us a few seconds later. 'Thunder created later' is wrong — both happen at the same instant. 'Sound faster than light' is the reverse of the truth. 'Eyes

react seconds faster' is false; the delay comes from the travel speeds of light and sound, not reaction times.

25. (A) Lightning is an enormous electric spark released when charges built up in the storm cloud suddenly discharge. 'Reflected sunlight' is wrong — lightning happens in dark storm clouds and even at night. 'Burning gas' is not the mechanism; it is an electrical discharge. 'Sound turning into light' is impossible — sound and light are different things, and the flash is not made from thunder.

26. (D) Light arrives almost instantly, so the gap is set by how long the slower sound takes to travel — a shorter gap means a shorter distance, so the storm is closer. 'Moving away with a shorter gap' reverses the relationship. 'Lightning getting weaker' confuses brightness with distance; the gap measures distance. 'Speed of light increasing' is false — that speed is fixed.

27. (B) When water vapour condenses it releases heat; this extra warmth makes the air rise even more, drops the central pressure further and draws in faster wind — a self-feeding loop. 'Cools and halts' is wrong — condensation releases heat, it does not cool the storm. 'Makes air heavier and sinks' is backwards; the warmed air rises. 'Raises pressure and slows wind' is the opposite of what intensifies a cyclone.

28. (D) The cycle begins with the Sun warming the sea so moist air rises (ii); that vapour condenses, releasing heat and dropping the pressure further (iii); the lowered pressure then sucks in faster wind (i), which fuels more rising air. Orders starting with (i) or (iii) put effects before the heating and rising that cause them. (ii)→(i)→(iii) wrongly has wind rushing in before the condensation that lowers the pressure to pull it.

29. (D) A cyclone runs on the heat and moisture of warm sea air. On land that supply is cut off, so the self-feeding cycle starves and the storm weakens. 'Colder than outer space' is irrelevant and false. 'Blocked by buildings' is far too weak to stop a storm hundreds of kilometres wide. 'Loses its eye instantly' is not the mechanism — the energy starvation, not the eye, is the cause.

30. (C) The very centre (eye) is calm and low-pressure; surrounding it the eye wall has the fiercest winds and heaviest rain; beyond that lie outer spiralling cloud bands. 'Violent eye, calm eye wall' swaps the two. 'High-pressure eye with no clouds' is wrong — the eye is a LOW-pressure core. 'Calm everywhere' contradicts the destructive winds a cyclone is famous for.

31. (C) The eye wall hits first with fierce wind; then the calm eye passes overhead; then the eye wall on the far side brings fierce wind again, now from the opposite direction because the storm has moved past. 'Steady mild wind' ignores the violent eye wall. 'No wind at all' is impossible inside a cyclone. 'Permanent calm at the end' is wrong — the far eye wall brings the wind back.

32. (C) Intensifying means a deeper low at the centre: the central pressure drops further, and because wind rushes from high pressure toward that ever-lower centre, the winds blow faster. 'Pressure rises and winds faster' is contradictory — rising pressure weakens a cyclone. 'Pressure falls but winds slower' breaks the link between a deeper low and faster wind. 'Both constant' describes a steady, not intensifying, storm.

33. (A) Storms and cyclones are low-pressure systems, so a sharp pressure drop warns that one may be approaching. 'Clear settled weather' goes with high or steady pressure, not a sharp fall. 'Wind about to stop' is wrong — falling pressure usually means rising wind. 'Temperature below freezing' is unrelated; a barometer measures pressure, not temperature.

34. (A) Wind blows from high to low pressure; the extreme low at the cyclone's centre pulls in surrounding air at great speed, and that inrush IS the cyclone's wind. 'Keeps the air still' is the opposite — the low pressure drives the wind. 'Raises sea level only far out at sea' is wrong; the surge strikes the coast. 'Dries and warms so no clouds form' contradicts the towering clouds of a cyclone.

35. (D) Cyclone, hurricane and typhoon are regional names for the same kind of swirling tropical low-pressure storm. 'Completely different events' is wrong — they share the same structure and cause. 'Fixed different sizes' is false; the names depend on region, not size. 'Old names for tornadoes' confuses tropical cyclones with the much smaller, different tornado.

36. (A) Hundreds of kilometres across, with a calm low-pressure eye ringed by violent winds and swirling bands — those are the defining features of a cyclone. A tornado is far narrower (a slim column, not hundreds of km wide). A land breeze is a gentle one-way coastal flow, not a swirling system. A 'thunderstorm with no rotation' lacks the swirling, eye-centred structure described.

37. (B) The storm surge — sea water heaped up by the low pressure and fierce winds and driven far inland — causes the worst flooding and loss of life on low coasts. 'Loud thunder' is only sound and harms no one. 'Hailstones' are minor compared with a surge. 'Lightning' is dangerous but kills far fewer than the surge's flooding.

38. (D) The cyclone's low pressure lets the sea rise and its strong winds pile the water into a surge that is pushed far inland, flooding low coasts. 'Slow rainwater' understates it — the surge is sea water driven by the storm, not gradual rain. 'Sea level drops' is the opposite of a surge. 'Entirely fresh river water' is wrong — a surge is sea water pushed ashore.

39. (C) Early warnings from satellites and radar give people time to evacuate to shelters, which has hugely cut deaths. 'Cyclones have become weaker' is false — they are as strong as ever. 'Sea no longer floods' is wrong; surges still occur, but people now get out of the way. 'Nobody lives near the coast' is untrue — coasts remain densely populated.

40. (B) Moving promptly to a safe shelter and following official guidance keeps you out of the surge and wind. 'Watching the waves at the beach' puts you in the path of the deadly surge. 'Ignoring it until the wind is strong' wastes the warning time meant for escape. 'Driving toward the sea' heads straight into the danger.

41. (C) Storing clean water and keeping supplies and a radio ready means you can survive flooding and follow updates. Opening all doors and windows lets the wind in to damage the house. Going out to photograph the storm exposes you to deadly winds and surge. Sheltering under a lone tree is dangerous in lightning and high wind.

42. (A) A cyclone's fierce winds and towering waves can easily capsize or sink small boats, so going to sea is deadly. 'Fish swim away' is a fishing concern, not the safety reason. 'Water too cold to sail in' is false and irrelevant to the danger. 'Sea water vanishes' is impossible.

43. (A) Fast-moving air has lower pressure; the express drags air past, lowering the pressure in the gap, so the higher-pressure air on the person's far side pushes them toward the train — which is why you must stand back. 'Moving train blows air that pushes you toward it' wrongly has the wind pushing you in. 'Magnetically attracts' is false. 'Rails vibrate and drag you' is not a real mechanism.

44. (D) The fast air in the stream is at lower pressure than the still air around it; if the ball drifts to the edge, the higher outside pressure pushes it back into the low-pressure stream, so it self-centres. 'Sticky air glues it' is not real. 'Becomes weightless' is false — its weight is still balanced by the upward airflow, not removed. 'Heat lifts by magnetism' confuses heat, lift and magnetism, none of which is the pressure effect at work.

45. (D) Air pressure pushes up on the card with more force than the small weight of the water pushes down, so the card and water stay put. 'Glued by being wet' is not what holds the weight up. 'Water can't fall upside down' is false — it would fall without the air pressure. 'Magnetic force' has nothing to do with water and glass.

46. (D) Air pressure comes from the weight of the air above; high on a mountain there is less air overhead, so the outside pressure is lower than at the seashore. 'More air above on the mountain' is backwards — there is less. 'No pressure at all' is false; the air is thinner, not absent. 'Exactly the same everywhere' ignores that pressure falls with height.

47. (B) Heated air expands and becomes less dense; the surrounding cooler, denser air sinks beneath it and pushes the lighter warm air up. 'Cools, shrinks, becomes heavier, so it rises' is self-contradictory — heavier air sinks. 'Air turns into water' is not what happens to heated air. 'Gains weight and heavy air rises' is doubly wrong: heating lightens air, and heavy air sinks.

48. (A) Persistent warming makes the air expand and rise away, so less air sits over the region and its pressure falls — that is why hot regions become low-pressure zones. 'Higher

pressure because warm air is heavier' is wrong; warm air is lighter and rises. 'No change' contradicts the link between heating and rising air. 'Zero pressure' is false — air remains, just less of it.

49. (B) Wind blows from high pressure to low pressure. By day the sea (cooler) is the higher-pressure side and the land (warm) is the lower-pressure side, so the breeze flows from sea to land. 'Land to sea' goes from low to high pressure, which is impossible. 'Straight upward only' ignores the horizontal pressure difference. 'In a circle with no net flow' contradicts the clear sea-to-land pressure gradient.

50. (C) As moist air rises and its vapour condenses, heat is released, the air rises more, and the central pressure drops further — so air keeps rushing in to the deepening low, a self-feeding loop. 'Wind manufactures low pressure with no heat or moisture' ignores the energy source that drives it. 'Vacuum cleaner unrelated to pressure' is wrong — it is precisely a pressure effect. 'Storm cools the sea and cool seas pull in wind' contradicts the warm-sea fuel a cyclone needs.

51. (B) Cooling makes the trapped air contract and its pressure fall; the outside air, still at higher pressure, then presses the bottle's sides inward, denting it. 'Freezer squeezed it mechanically' is not what a freezer does. 'Cold made the plastic expand and fold' is wrong — cold contracts plastic, and the effect here is about the air's pressure. 'Water soaked through' is false; a sealed bottle does not let freezer water in, and that would not dent it.

52. (A) Air moves from high pressure to low pressure, so the high-pressure region feeds air outward toward the low. 'Stays trapped' contradicts the pressure difference that drives wind. 'Flows inward from low to high' is the reverse of the rule. 'Rises straight up leaving the surface empty' is not what a high-pressure region does — high pressure is generally associated with sinking, not a surface emptied of air.

53. (C) When warm air rises and moves off, less air remains over the surface, so the weight pressing down — the pressure — falls. 'Leaves more air behind' is the opposite of air moving away. 'Sun adds pressure to cool surfaces' is unrelated to this mechanism. 'Turns into water that weighs the surface down' is wrong — the air rising away is what lowers the pressure.

54. (A) Fast-moving fluid has lower pressure; with low pressure between them and higher pressure outside, the objects are pushed together — the same reason close-sailing ships can be drawn in. 'Moving fluids exert higher pressure' is the reverse of the truth. 'Carries them by sticking' is not the pressure mechanism. 'Magnetic force' has nothing to do with moving air or water.

55. (C) A deeper central low means a bigger pressure difference between the surroundings and the centre, so air rushes in faster — faster winds. 'Lower pressure calms the air' is

backwards. 'No wind' is wrong — the deep low drives strong inflow. 'Wind blows outward from the centre' reverses the direction; air rushes IN toward the low.

56. (D) Heating expands the air in the flask, which pushes on the water and lowers its level; cooling contracts the air and the water rises again — direct evidence that air expands on heating and contracts on cooling. 'Created and destroyed' is false; the same air just changes volume. 'Always pushes water up' ignores that the level fell when heated. 'No response to temperature' contradicts the observed movement.

57. (D) The true driver is pressure: wind flows from high to low. Because hot regions tend to become low-pressure, the flow often runs from cooler (higher-pressure) toward warmer (lower-pressure) regions — so the student's pattern is roughly right but for the wrong stated reason. 'Cold air attracted to heat' wrongly makes temperature the cause. 'Hot to cold in every case' is too absolute and ignores pressure. 'Nothing to do with pressure' denies the chapter's central idea.

58. (A) Boiling replaces the air with steam; on cooling the steam condenses to a few drops, so almost no gas is left and the inside pressure collapses far more than merely cooling trapped air would — giving a dramatic crush. 'Boiling softens the metal' is not the reason; the effect is about pressure. 'Room-temperature air cannot be cooled' is false. 'Boiling adds extra air' is wrong — it drives air OUT and replaces it with steam.

59. (A) Crushing needs a low inside pressure compared with outside; an open hole lets outside air rush in to equalise the pressure as the steam condenses, so no difference builds and the can stays whole. 'Crush even faster' is wrong — an open can cannot build the needed pressure difference. 'Metal melts' and 'steam catches fire' are not what happens when a can cools with a hole in it.

60. (C) As the cyclone moves over the town, the front eye wall hits with fierce wind, the calm eye passes over (the lull), then the rear eye wall brings fierce wind from the other direction. 'Stops and restarts itself' is not how a moving storm behaves. 'Two separate cyclones' is unnecessary — one storm's structure explains it. 'Changes its mind for no reason' ignores the eye-and-eye-wall structure.